Prime-Force Correspondence: Gauge Groups from Ramified Coverings of $\mathrm{Spec}(\mathbb{Z})$

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We observe that the three gauge group factors of the Standard Model admit a uniform description in terms of number field extensions of $\mathbb{Q}$ ramified at a single prime: U(1) $\leftrightarrow$ $\mathbb{Q}$ (trivial), SU(2) $\leftrightarrow$ $\mathbb{Q}(\sqrt{2})$ (ramified at p=2), SU(3) $\leftrightarrow$ $\mathbb{Q}(\zeta_9)^+$ (ramified at p=3). The pattern is SU(N) $\leftrightarrow$ degree-N covering with Galois group $\mathbb{Z}/N$, ramified at p=N. Two independent numerical coincidences support this correspondence: $\cos\theta_W = \mathrm{reg}(\mathbb{Q}(\sqrt{2})) = \log(1+\sqrt{2})$ agrees with m_W/m_Z to 0.001%, and $10\,h{\cdot}R(\mathbb{Q}(\zeta_9)^+) = 8.493$ agrees with $\alpha_s(M_Z)^{-1} = 8.482$ to 0.13%. We propose that gauge symmetry breaking is encoded in the ramification structure of arithmetic coverings, with the Frobenius element playing the role of gauge holonomy. The correspondence makes a falsifiable prediction for m_W at sub-MeV precision.

OBSERVATION

The Standard Model gauge group $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ has resisted derivation from a deeper principle. We report a numerical observation that may point toward such a principle.

Consider the following number fields:

- $\mathbb{Q}$ itself (degree 1 over $\mathbb{Q}$).
- $K_2 = \mathbb{Q}(\sqrt{2})$, the simplest real quadratic field. Discriminant $\Delta = 8 = 2^3$; ramified at $p = 2$ only; class number $h = 1$; regulator $R = \log(1+\sqrt{2}) = 0.88137\ldots$
- $K_3 = \mathbb{Q}(\zeta_9)^+ = \mathbb{Q}(\cos 2\pi/9)$, the maximal real subfield of the 9th cyclotomic field. Degree 3; discriminant $\Delta = 81 = 3^4$; ramified at $p = 3$ only; Galois group $\mathbb{Z}/3$; $hR = 0.84931\ldots$

Each field is the *minimal* extension of $\mathbb{Q}$ with cyclic Galois group $\mathbb{Z}/N$ that is ramified at $p = N$ *only*, for $N = 1, 2, 3$ respectively.

We observe:

$$\cos\theta_W = \frac{m_W}{m_Z} \stackrel{?}{=} \mathrm{reg}(K_2) = \log(1+\sqrt{2}). \tag{1}$$

Numerically: $\log(1+\sqrt{2}) = 0.881374$ vs. $m_W/m_Z = 0.881361 \pm 0.000147$. Agreement: 0.09σ.

$$\frac{1}{\alpha_s(M_Z)} \stackrel{?}{=} 10\; hR(K_3) = 10 \times 0.84931. \tag{2}$$

Numerically: 8.493 vs. $1/\alpha_s = 8.482 \pm 0.008$. Agreement: 0.13%.

THE CORRESPONDENCE

Gauge	Field	Degree	Gal	Ram. at	Invariant
U(1)	$\mathbb{Q}$	1	$\{1\}$	—	α
SU(2)	K_2	2	$\mathbb{Z}/2$	p=2	$\cos\theta_W$
SU(3)	K_3	3	$\mathbb{Z}/3$	p=3	α_s^{-1}

The pattern: SU(N) $\leftrightarrow$ degree-N cyclic extension of $\mathbb{Q}$, ramified at $p = N$ only, with Galois group $\mathbb{Z}/N$.

Why $p = 2$ and weak force? The weak interaction is the *only* force that violates parity (P) [1]. Parity is a $\mathbb{Z}/2$ symmetry, and $p = 2$ is the unique prime governing the even/odd distinction. $K_2 = \mathbb{Q}(\sqrt{2})$ is the unique class-number-one real quadratic field ramified at $p = 2$ only. The chain P-violation $\leftrightarrow$ $\mathbb{Z}/2$ $\leftrightarrow$ p=2 $\leftrightarrow$ K_2 provides a structural (if not yet rigorous) link between the weak force and the arithmetic of $\mathrm{Spec}(\mathbb{Z})$.

Why degree 3 for SU(3)? A field ramified at $p = 3$ only *cannot* be quadratic: for $d = 3$, the discriminant is $\Delta = 12 = 4 \times 3$, which is ramified at both $p = 2$ and $p = 3$. This is because $3 \equiv 3 \pmod 4$; for any $p \equiv 3 \pmod 4$, the quadratic field $\mathbb{Q}(\sqrt{p})$ has $\Delta = 4p$, inevitably introducing $p = 2$. A "pure" covering ramified at $p = 3$ only requires degree ≥ 3. The minimal such is the cyclic cubic $K_3 = \mathbb{Q}(\zeta_9)^+$, with $\Delta = 3^4$. Thus *number theory itself forces* SU(3) to live in a higher-degree covering than SU(2).

THE CLASS NUMBER FORMULA CONNECTION

For K_2, the Dirichlet class number formula gives:

$$L(\chi_8, 1) = \frac{hR}{\sqrt{\Delta}} = \frac{\log(1+\sqrt{2})}{\sqrt{8}}. \tag{3}$$

Since $\cos\theta_W = hR = \log(1+\sqrt{2})$ and $h = 1$:

$$\cos\theta_W = L(\chi_8, 1)\,\sqrt{\Delta(K_2)}. \tag{4}$$

For K_3, the class number formula gives hR in terms of $|L(\chi_3, 1)|^2$ where χ_3 is a character of order 3 mod 9:

$$hR(K_3) = \tfrac{9}{4}\,|L(\chi_3, 1)|^2. \tag{5}$$

In both cases, the physical parameter (mixing angle or coupling) equals an arithmetic invariant that appears in the class number formula—one of the deepest results in algebraic number theory [2].

The Dedekind zeta function of K factorizes as $\zeta_K(s) = \zeta(s) \cdot L(\chi, s) \cdots$, separating into the "base" zeta (governing α and G) and the L-factors (governing mixing angles and non-abelian couplings). At $s = 1$, $\zeta(s)$ has a pole (gravity), while each L-factor is finite (mixing parameters).

FROBENIUS AS HOLONOMY

In the covering $K_2/\mathbb{Q}$, each unramified prime $p \neq 2$ carries a Frobenius element $\mathrm{Frob}_p \in \mathrm{Gal}(K_2/\mathbb{Q}) = \mathbb{Z}/2$:

Prime type	Frob_p	Physical analog
Split ($p \equiv \pm 1 \pmod 8$)	id	trivial holonomy
Inert ($p \equiv \pm 3 \pmod 8$)	σ	nontrivial holonomy
Ramified ($p = 2$)	undefined	source of the force

In gauge theory, the holonomy of a connection around a loop classifies the local gauge transformation. We propose that the Frobenius element at each prime p plays the role of holonomy of an arithmetic gauge connection, with the ramification locus ($p = 2$ for the weak force) as the "source" of the gauge field.

PREDICTIONS

1. $m_W = m_Z \times \log(1+\sqrt{2}) = 80.370$ GeV. Current experimental value: 80.369 ± 0.013 GeV (PDG 2024), 0.09σ from prediction. Future measurements at sub-MeV precision (FCC-ee, CEPC) will provide a decisive test.

2. $\sin^2\theta_W = 1 - \log^2(1+\sqrt{2}) = 0.22318$, to be compared with the on-shell value 0.22320 ± 0.00004.

3. The strong coupling is related to K_3 via $\alpha_s^{-1}(M_Z) = c \cdot hR(K_3)$ with $c = 10$ (the origin of this factor is currently unexplained).

WHAT THIS IS AND WHAT IT IS NOT

This paper reports a *numerical observation* and a *structural correspondence*, not a derivation. We do not derive the Standard Model gauge group from number theory; we observe that the group structure matches the ramification structure of specific number field extensions in a way that produces correct physical parameters.

The key weakness is the absence of a dynamical mechanism: we do not explain *why* gauge couplings should equal arithmetic regulators. The factor of 10 in the α_s relation is unexplained. The correspondence does not extend to CKM or PMNS mixing matrices.

The key strength is falsifiability: Eq. (1) predicts m_W to 0.1 MeV, well within reach of next-generation colliders. If m_W deviates from $m_Z \log(1+\sqrt{2})$ by more than a few MeV, the correspondence is ruled out.

The correspondence $\mathrm{SU}(N) \leftrightarrow$ ramification at $p = N$ is forced by the structure of algebraic number theory (the impossibility of a quadratic field ramified at $p = 3$ only). Whether nature respects this arithmetic constraint is an empirical question.

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[3] J. Neukirch, *Algebraic Number Theory* (Springer, 1999).
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